PS - 0

1. Given, $\nabla f(x) = \begin{bmatrix} \frac{\partial}{\partial x_1} f(x) \\ \vdots \\ \frac{\partial}{\partial x_n} f(x) \end{bmatrix}$ where $x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$

Hessian $\nabla^2 f(x)$ of a function $f: R^n \to R$ is the $n \times n$ ~~system~~, symmetric matrix of twice partial derivatives,

$$\nabla^2 f(x) = \begin{bmatrix} \frac{\partial^2}{\partial x_1^2} f(x) & \frac{\partial^2}{\partial x_1 \partial x_2} f(x) & \cdots & \frac{\partial^2}{\partial x_1 \partial x_n} f(x) \\ \frac{\partial^2}{\partial x_2 \partial x_1} f(x) & \frac{\partial^2}{\partial x_2^2} f(x) & \cdots & \frac{\partial^2}{\partial x_2 \partial x_n} f(x) \\ \vdots & & & \\ \frac{\partial^2}{\partial x_n \partial x_1} f(x) & \frac{\partial^2}{\partial x_n \partial x_2} f(x) & \cdots & \frac{\partial^2}{\partial x_n^2} f(x) \end{bmatrix}$$

(a) Let $f(x) = \frac{1}{2} x^T A x + b^T x$ (A is symm matrix, $b \in R^n \to$ vector)

$\nabla f(x) = ?$

$\nabla f(x) = \frac{1}{2} x^T A x + bx \quad [\because b^T = b,$

here we using scalar calculus sice $x^T A x$ is scalar $\to$ treating it as scalar mul

$= \frac{1}{2} A x^2 + bx$ $\quad$ ~~$x^T = x$~~ $\quad$ (Never write this wrong)

$\therefore \nabla f(x) = \frac{1}{2} \times 2 x \times A + b$

$= Ax + b$

$\to x^T A x = A x^2$ [$x^T A x$ is matrix analog of ax^2 in scalar case]

(b) Let $f(x) = g(h(x))$, where $g: R \to R$ is differentiable, $h: R^n \to R$ is differentiable.

what is $\nabla f(x)$?

$\Rightarrow$

$\nabla g(h(x)) = g'(h(x)) \cdot h'(x)$ $\quad \nabla h(x)$

$\Rightarrow \frac{\partial g(h(x))}{\partial x_i} = \frac{\partial g(h(x))}{\partial h(x)} \cdot \frac{\partial h(x)}{\partial x_i}$

$= \underbrace{g'(h(x))}_{\text{scalar}} \cdot \underbrace{\nabla h(x)}_{\text{vector as } h(x) \text{ is scalar}}$

(c) Let $f(x) = \frac{1}{2} x^T A x + b^T x$ ($A \to$ symm, $b \to \mathbb{R}^n$)

$\nabla^2 f(x) = ?$

$f(x) = \frac{1}{2} x A x + bx$

$= \frac{1}{2} A x^2 + bx$

$\nabla f(x) = f'(x) = Ax + b$ (as we get from (a))

$\nabla^2 f(x) = f''(x) = A = \begin{bmatrix} A_{11} & A_{12} & & A_{1n} \\ A_{21} & A_{22} & \cdots & \\ A_{31} & & A_{33} & \\ \vdots & & & A_{nn} \\ A_{n1} & \cdots & & \end{bmatrix}$

$= \frac{\partial (Ax+b)}{\partial x_1}$

$\frac{\partial}{\partial x_1}(Ax+b) + \frac{\partial}{\partial x_2}(Ax+b) + \cdots$

Another representation $\Rightarrow$

$$\nabla^2 f(x) = \begin{bmatrix} \frac{\partial \nabla f(x)}{\partial x_1} & \frac{\partial \nabla f(x)}{\partial x_2} & \cdots & \frac{\partial \nabla f(x)}{\partial x_n} \\ \vdots & & & \vdots \\ \vdots & & & \end{bmatrix}$$

$$= \begin{bmatrix} \frac{\partial (Ax+b)}{\partial x_1} & \frac{\partial (Ax+b)}{\partial x_2} & \cdots & \frac{\partial (Ax+b)}{\partial x_n} \\ \\ \end{bmatrix}$$

$= A$

(d) Let $f(x) = g(a^T x)$, $g : \mathbb{R} \to \mathbb{R}$ continuously differentiable & $a \in \mathbb{R}^n$ is a...

$\nabla f(x) = ?$ $\nabla^2 f(x) = ?$

$\nabla f(x) \Rightarrow$ $\nabla f(x) = \nabla g(a^T x)$

$= g'(a^T x) \cdot \frac{\partial (a^T x)}{\partial x}$

$= g'(a^T x) \cdot a^T$

$= g'(a^T x) \cdot a$ [$a^T = a$] as it is a vector quantity

(c) Let $f(x) = \frac{1}{2} x^T A x + b^T x$ ($A \to$ sym
$b \to \mathbb{R}^n$ vec)

$\nabla^2 f(x) = ?$

$f(x) = \frac{1}{2} x A x + b x$

$= \frac{1}{2} A x^2 + b x$

$\nabla f(x) = f'(x) = Ax + b$ (we get it from (a))

$\nabla^2 f(x) = f''(x) = A$ (not good notation for this)

$= \frac{\partial (Ax+b)}{\partial x}$

$\frac{\partial}{\partial x_1}(Ax_1 + b) + \frac{\partial}{\partial x_2}(Ax_2+b) + \cdots$

$$= \begin{bmatrix} A_{11} & A_{12} & \cdots & A_{1n} \\ A_{21} & A_{22} & \cdots & \\ A_{31} & & A_{33} & \\ \vdots & & & A_{nn} \\ A_{n1} & & & \end{bmatrix}$$

Another representation $\Rightarrow$

$$\nabla^2 f(x) = \begin{bmatrix} \frac{\partial \nabla f(x)}{\partial x_1} & \frac{\partial \nabla f(x)}{\partial x_2} & \cdots & \frac{\partial \nabla f(x)}{\partial x_n} \\ \vdots & & & \vdots \end{bmatrix}$$

$$= \begin{bmatrix} \frac{\partial (Ax+b)}{\partial x_1} & \frac{\partial (Ax+b)}{\partial x_2} & \cdots & \frac{\partial (Ax+b)}{\partial x_n} \end{bmatrix}$$

$= A$

(d) Let $f(x) = g(a^T x)$, $g : \mathbb{R} \to \mathbb{R}$ continuously differentiable & $a \in \mathbb{R}^n$ is vec.

$\nabla f(x) = ?$ $\nabla^2 f(x) = ?$

$\nabla f(x) = \nabla g(a^T x)$

$= g'(a^T x) \cdot \frac{\partial (a^T x)}{\partial x}$

$= g'(a^T x) \cdot a^T$

$= g'(a^T x) \cdot a$ $[a^T = a]$ as it is a vector quantity

$\nabla^2 f(x) = \nabla^2 g(a^T x)$

$= \dfrac{\partial^2 g(a^T x)}{\partial x_i \partial x_j}$

Another -> $\dfrac{\partial}{\partial x} g'(a^T x) \cdot a$

let's take. $h(x) = a^T x$

$\therefore \dfrac{\partial^2 g(h(x))}{\partial x_i \partial x_j}$

$= \dfrac{\partial^2 g(h(x))}{\partial (h(x))^2} \dfrac{\partial (h(x))}{\partial x_i} \dfrac{\partial (h(x))}{\partial x_j}$

$\dfrac{\partial}{\partial x_i} g'(a^T x) \cdot a_i = g''(a^T x) \cdot a_j \cdot a_i = g''(a^T x)\, a a^T$

$= g''(a^T x) \dfrac{\partial (h(x))}{\partial x_i} \cdot \dfrac{\partial (h(x))}{\partial x_j}$

~~$g'' h(a^T x) =$~~

$\dfrac{\partial^2 g(a^T x)}{\partial x_i \partial x_j} = g''(a^T x) \cdot a_i \cdot a_j \quad \dfrac{\partial (a^T x)}{\partial x_i} \cdot \dfrac{\partial (a^T x)}{\partial x_j}$

$= g''(a^T x) \cdot a_i \cdot a_j$

$= g''(a^T x) \begin{bmatrix} a_1 a_1 & a_1 a_2 & a_1 a_n \\ a_2 a_1 & a_2 a_2 & a_2 a_n \\ \vdots & & \vdots \\ a_n a_1 & a_n a_2 & a_n a_n \end{bmatrix}$

$= g''(a^T x) \cdot a a^T \quad \left[\begin{matrix} a^T = a \\ a_j = a^T \\ a_i = a \end{matrix} \right]$

Error Correction :=>

(a) $x^T A x \rightarrow x^T \cdot A \cdot x$ (Product of 3)

when we are diff -> wrt x_k -> x appears twice so we get 2 term like=> $\dfrac{d}{dx}(x \cdot f(x))$

$= \dfrac{d}{dx} x f(x) \cdot \dfrac{d}{dx}(x)$

$= f(x) + x \cdot f'(x) \mid + \dfrac{d}{dx} x \cdot \dfrac{d}{dx} f(x)$

Term 1 :→

$x^T A x = \sum_i \sum_j x_i A_{ij} x_j$

i → fix ...... diff. w.r.t. x_k where $j = k$

$\therefore \frac{\partial}{\partial x_k}(x^T A x) = \sum_i x_i A_{ik}$ ↳ k^{th} ele of $A^T x$

$(A^T x)_k = \sum_i (A^T)_{ki} x_i = \sum A_{ik} x_i = \sum x_i$

Term 2 :→

j → fix ...... diff w.r.t. x_k where $i = k$

$\frac{\partial}{\partial x_k}(x^T A x) = \sum_j A_{ij} x_j$ ↳ k^{th} ele of Ax

Adding Both :→

$\frac{\partial}{\partial x_k}(x^T A x) = (Ax)_k + (A^T x)_k$

(→ in vector → $\nabla(x^T A x) = Ax + A^T x$

$= 2Ax$

(A sym → $A^T = A$) →

Another term →

$b^T x$

→ $\nabla(B^T x) = Bx$

$\frac{\partial}{\partial x_k}(b^T x) = \underline{b^T e_k}$ [b is vector, $b \to \sim$]

$= \frac{\partial}{\partial x_k}(\sum_i b_i x_i)$

$= b_k$ (only $i = k$ term surviving other b_i are constants)

$f(x) = \frac{1}{2} x^T A x + b^T x$

$\nabla f(x) = \frac{1}{2} \cdot 2Ax + b_k$

$= Ax + b_k$ (Ans)

2: Positive definite matrices $\Rightarrow$

(a) Let $z \in R^n$ be an n-vector. Show that $A = zz^T$ is positive semidefinite.

$\Rightarrow$ $A = A^T$ (one cond'n)

$A^T = (zz^T)^T = zz^T = A$ (satisfied) $\quad ((B^T)^T = B)$

$x^T A x \geq 0$ (another cond'n) $\quad$ [formula: $(AB)^T = B^T A^T$]

$x^T A x = x^T zz^T x = x^T z \cdot (x^T z)^T$

$= (x^T z)^2 \geq 0$

It's a scaler

As we know whole sq can't be (-ve)

shape/size $\Rightarrow$

$A = n \times 1$

$A^T = 1 \times n$

$z = n \times 1$

$\therefore A^T z = (1 \times n) \cdot (n \times 1)$

$= (1 \times 1) \to$ scalar

(b) Let $z \in R^n$ be a non-zero n-vector. Let $A = zz^T$. What is the null-space of A? Rank of A?

$\Rightarrow$ $z \to$ shape $(n \times 1)$

For positive Definite (PD) $\to$ null space only $\{0\}$, condn $x^T A x > 0$ $\forall x \neq 0$

for positive Semi Definite (P.S.D) $\to$ contains non-zero vectors, condn $x^T A x = 0$ for some $x \neq 0$

$\to$ we have proved

$A = zz^T$ is positive semi definite

$x^T A x = (x^T z)^2 = 0$ $\to$ if here x is a perpendicular vector of z then $\boxed{x^T z = 0}$

follow rule of P.S.D

$x^T A x$ $\therefore A = zz^T$

$\Rightarrow \boxed{Az = z}$

$\boxed{\therefore x^T A x = (x^T z) = 0}$

$x^T A x = x^T z z^T x$

$= \underbrace{x^T z \cdot (x^T z)^T}_{\text{scaler}}$

$= (x^T z)^2$

$= 0 \quad [\because x^T z = 0]$

D Rank of A is 1 as $A = z z^T$ and z^T getting multiplied with all z_i (z vector).

(c)

$A \in R^{n \times n} \Rightarrow$ p.s.d

$\therefore A \geq 0$

$A = A^T$

$x^T A x \geq 0 \quad \forall x \in R^n$

(I) $B A \cdot B^T \geq 0$ ~~$= B A^T B^T = B \cdot (B \cdot A)^T$~~

$[\because x^T A x \geq 0]$

(satisfied)

(II) $(B A B^T)^T = (B^T)^T \cdot A^T \cdot B^T$

$= B A^T B^T$

$= B A B^T \quad [\because A^T = A]$

$(A^T = A)$ (satisfied)

(III) $x^T (B A B^T) x = (x^T B) A (B^T x)$

$= \underbrace{(x^T B) A (x^T B)^T}_{\text{scaler}} \geq 0$

$= (x^T B)^2 \cdot A \geq 0$

(satisfied)

$\therefore$ Hence BAB^T is P.S.D.

3. (a) $A \in \mathbb{R}^{n \times n}$; $A = TΛT^{-1}$; $T \in \mathbb{R}^{n \times n}$; $Λ = \text{diag}(λ_1 \ldots λ_n)$
matrix

$T = [t^{(1)} \ldots t^{(n)}]$; $t^{(i)} \in \mathbb{R}^n$

$A = TΛT^{-1}$

$\Rightarrow AT = TΛ$

$\Rightarrow \underset{\text{matrix}}{A} \cdot \underset{\text{matrix}}{[t^{(1)} \ldots t^{(n)}]} = [t^{(1)} \ldots t^{(n)}] \cdot \begin{bmatrix} λ_1 & 0 & 0 & \cdots & 0 \\ 0 & λ_2 & 0 & \cdots & 0 \\ \vdots & & & & \vdots \\ 0 & 0 & 0 & \cdots & λ_n \end{bmatrix}$

vector — diag mat

(taking i th element)

$\Rightarrow A \cdot t^{(i)} = (t^i) \cdot λ_i =$

$\Rightarrow A \cdot t^{(i)} = λ_i (t^{(i)})$ as t^i is vector

(Proved)

$\boxed{\text{Another}} \Rightarrow U \in \mathbb{R}^{n \times n}$, $U^T U = I$, $A \in \mathbb{R}^{n \times n} \to \text{sym}(A^T = A)$

orthogonal mat — matrix

diagonal mat

Prove $U^T A U = Λ$ or, $A = UΛU^T$

orthogonal mat

$\to A^T A = AA^T = I$

$\to \boxed{A^T = A^{-1}}$

$\to AA^T = I$

$\Rightarrow AU = \begin{bmatrix} λ_1 & 0 & 0 & \cdots & 0 \\ 0 & λ_2 & 0 & \cdots & 0 \\ \vdots & & & & \\ 0 & 0 & 0 & \cdots & λ_n \end{bmatrix} \cdot \begin{bmatrix} U_{11} & U_{12} & \cdots & U_{1n} \\ U_{21} & U_{22} & & \vdots \\ U_{n1} & & \cdots & U_{nn} \end{bmatrix}$

$(n \times n)$, $(n \times n)$

$= \begin{bmatrix} λ_1 U_{11} & & \\ & λ_2 U_{22} & \\ & & λ_n U_{nn} \end{bmatrix}$

$(1 \times n)$

$= [λ_1 U_{11} \; λ_2 U_{22} \ldots λ_n U_{nn}]$

$(1 \times n)$

$U^T(AU)$

$= AU(U^T) = \begin{bmatrix} λ_1 U_{11} & & \\ & λ_2 U_{22} & \\ & & λ_n U_{nn} \end{bmatrix} \begin{bmatrix} U^T_{11} & U^T_{12} & \cdots & U^T_{1n} \\ U^T_{21} & U^T_{22} & & \\ U^T_{n1} & & & U^T_{nn} \end{bmatrix}$

$(1 \times n)$ $(n \times n)$

scalar

$\left(\begin{matrix} U_{11} \cdot U^T_{11} = I \\ U_{22} \cdot U^T_{22} = I \end{matrix} \right)$ scalar

$= \begin{bmatrix} λ_1 U_{11} \cdot U^T_{11} & & \\ & λ_2 U_{22} U^T_{22} & \\ & & λ_n U_{nn} U^T_{nn} \end{bmatrix} = I \begin{bmatrix} λ_1 & 0 & 0 & \cdots & 0 \\ 0 & λ_2 & & & \\ 0 & & & & λ_n \end{bmatrix} = IΛ$

scaler

$\therefore U^T A U = (AU) \cdot U^T = \underbrace{I}_{\text{scaler (value}=1)} A = \cancel{A} \cdot 1 A = A$ (Proved)

(b) $U = [u^{(1)} \ldots u^{(n)}]$, $u^{(i)} \in \mathbb{R}^n$, $A = U \Lambda U^T$

Prove → $u^{(i)}$ → eigen vector of A

& $A(u)^{(i)} = \lambda_i u^{(i)}$

, $\Lambda = \text{diag}(\lambda_1, \ldots \lambda_n)$

$= \begin{bmatrix} \lambda_1 & 0 & 0 & \cdots & 0 \\ 0 & \lambda_2 & & \cdots & 0 \\ 0 & 0 & \cdots & & \lambda_n \end{bmatrix}$

⇒

(b) $A = U \Lambda U^T$

$\Rightarrow AU = \Lambda U \lambda = \lambda U$

must treat

↳ taking ith element

as $\lambda_i \cdot u^i$ are now scalers, we can just implement commutative rules.

$\boxed{A(u)^{(i)} = \lambda_i \cdot (u)^{(i)}}$ (Proved)

(Proved) λ_i = i th diag ele

so, $u^{(i)}$ is eigen value of A (Proved)

(c)

Prove → A is P.S.D then $\lambda_i(A) \geq 0$ for each 'i'

⇒

given, A is P.S.D ($A \in \mathbb{R}^{n \times n}$ matrix)

$\therefore A \geq 0$, $A = A^T$, $x^T A x \geq 0$

, $x \in \mathbb{R}^n$ → vector

$\lambda_i A \cancel{=} \cancel{\lambda_i U \Lambda U^T}$

↳ if $\lambda_i \geq 0$ $\cancel{\lambda AU \Lambda U^T}$

$\lambda_i A \geq 0$ will be proved as $A \geq 0$ already proved

from (a) we know $A(t)^{(i)} = \lambda_i t^{(i)}$

$\Rightarrow \cancel{(t^{(i)})^T \cdot A \cdot t^{(i)} = \lambda_i \|t^{(i)}\|_2 \Rightarrow \lambda_i \geq 0}$

$\because \underline{(t^{(i)})^T \cdot A \cdot (t)^{(i)}} = \underbrace{(t^{(i)})^T \cdot \lambda_i \cdot t^{(i)}}_{\text{scaler}}$ [A diag $(t^{(i)})^T$ both side]

$\Rightarrow \quad \lambda_i \quad = \lambda_i \underbrace{(t^{(i)})^T \cdot t^{(i)}}_{\text{just dot prod of } t^{(i)}}$

$= \lambda_i \|t^{(i)}\|_2^2$

$\cancel{\Rightarrow \lambda_i \geq 0}$

Now,

$\cancel{\lambda_i \|t^{(i)}\|_2^2}$ $\boxed{x^T A x \geq 0 \ \forall x}$ (known)

$\Rightarrow (t^{(i)})^T A \cdot (t^{(i)}) \geq 0$

$\Rightarrow \lambda_i \|t^{(i)}\|_2^2 \geq 0$

$\Rightarrow \boxed{\|t^{(i)}\|_2^2 > 0}$ ($\because t^{(i)} \to$ eigenvalue $\to$ known $t^{(i)} \neq 0$ given)

$\Rightarrow \boxed{\lambda_i \geq 0}$ [dividing by $\|t^{(i)}\|_2^2$ both sides]

(Proved)

$\therefore$ for each $\boxed{\lambda_i \ A \geq 0}$ (Proved)